



ECEN 314 - Signals and Systems

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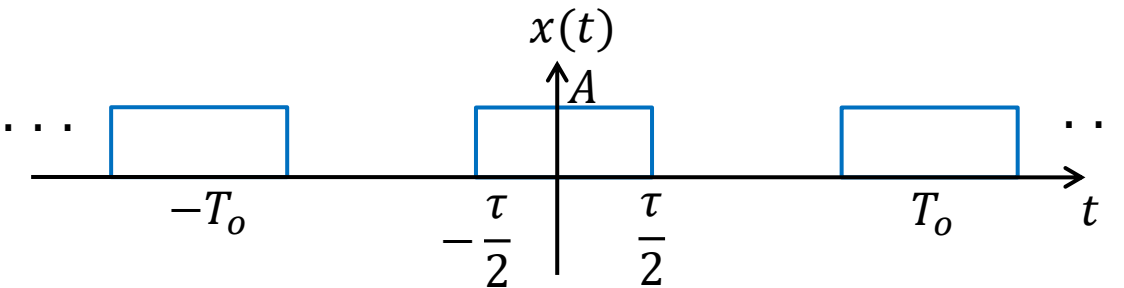
Module 11 - Fourier Transforms – Part I

Learning Objectives



- Be familiar with the definition of Fourier transform and the inverse Fourier transform.
- Be familiar with various properties of Fourier transforms.
- Be able to find the Fourier transform of common signals:
 - Square pulses
 - Triangular pulses
 - Exponentials (both one and two-sided)
- Be able to use properties of Fourier transforms to find Fourier transforms of more complicated signals.
- Be able to use table of Laplace transforms to find Fourier transforms for one-sided signals.

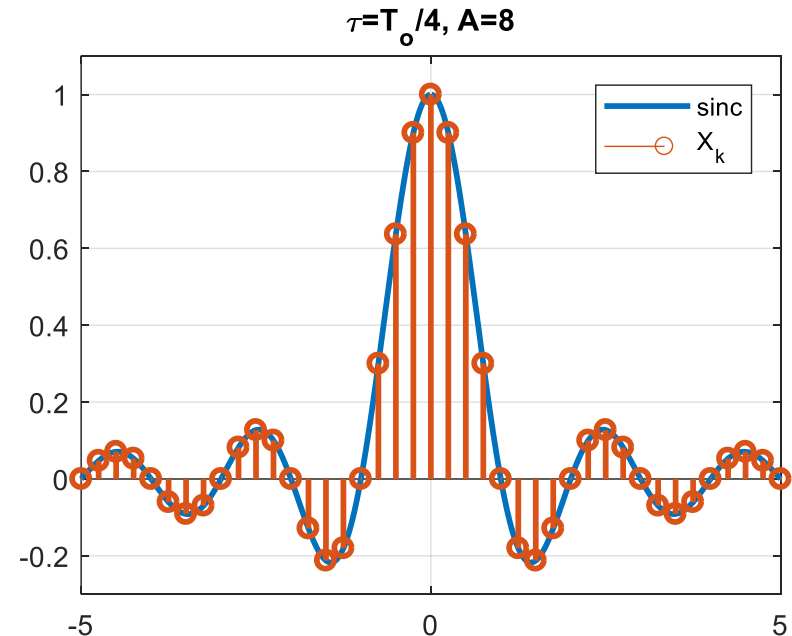
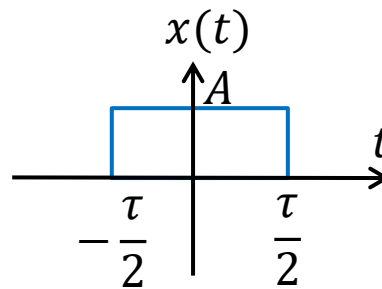
Square Wave $\rightarrow$ Square Pulse



$$X_k = \frac{A\tau}{2T_o} \text{sinc}\left(\frac{k\tau}{2T_o}\right)$$

Consider what happens to the periodic square wave in both the time and frequency domain as we allow $T_o \rightarrow \infty$.

- ✓ In the time domain, the square wave becomes an aperiodic square pulse.



- ✓ In the frequency domain,
 - $\omega_o \rightarrow 0$, so that the FS coefficients become more and more densely spaced, until the frequency content becomes continuous,
 - $X_k \rightarrow 0$ for all k , so that each individual FS coefficient gets smaller and smaller, until the strength of each frequency component becomes zero.

Frequency Content of Aperiodic Signals



$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_o t} \leftrightarrow X_m = \frac{1}{T_o} \int_{T_o} x(t) e^{-jm\omega_o t} dt$$

In order to avoid the indeterminate expressions, we will pass to the limit as $T_o \rightarrow \infty$, by defining:

$$\begin{aligned} \omega_o &\rightarrow \Delta\omega, \text{ or } f_o \rightarrow \Delta f, \\ k\omega_o &\rightarrow \omega, \text{ or } kf_o \rightarrow f, \\ X(f) &= \frac{X_k}{\Delta f} = T_o X_k, \text{ or } X(\omega) = \frac{2\pi X_k}{\Delta\omega} \end{aligned}$$

With these definitions

$$\begin{aligned} X(f) &= X_k T_o \rightarrow \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt \quad \text{or} \quad X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \\ x(t) &= \sum_{k=-\infty}^{\infty} X_k e^{j2\pi k f_o t} = \sum_{k=-\infty}^{\infty} X(k f_o) e^{j2\pi k f_o t} \Delta f \rightarrow \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df \\ &\quad \text{or} \\ x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \end{aligned}$$

Now $X(f)$ (or $X(\omega)$) is to be interpreted as the density of the signal content per unit frequency. It is known as the *Fourier transform* of the signal $x(t)$.

Fourier Transform



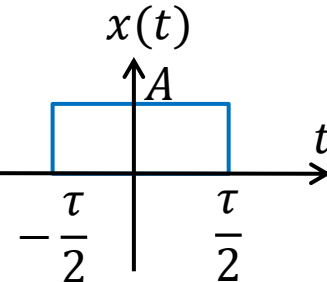
For aperiodic signals, we use the *Fourier transform* to describe the frequency content of the signal,

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \leftrightarrow X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

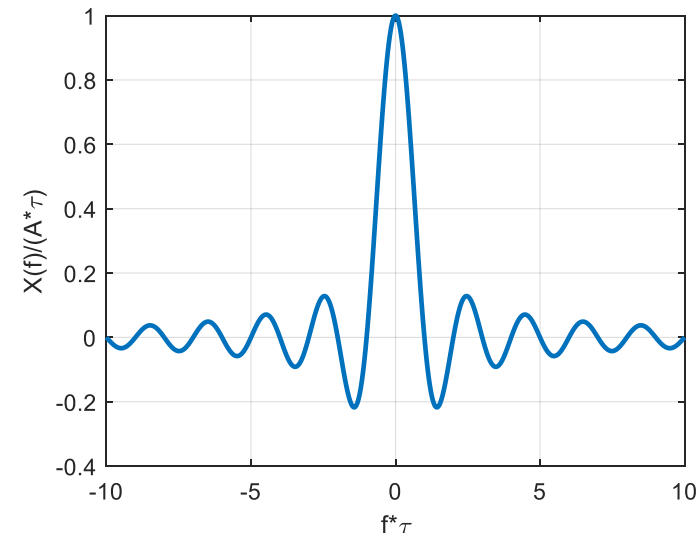
or with frequency measured in Hz rather than rad/sec

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df \leftrightarrow X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$$

Example:

$$x(t) = A \cdot \text{rect}(t/\tau) = \begin{cases} A, & |t| < \tau/2, \\ 0, & |t| > \tau/2. \end{cases}$$


$$\begin{aligned} X(f) &= \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt = A \int_{-\tau/2}^{\tau/2} e^{-j2\pi f t} dt \\ &= A \frac{e^{-j2\pi f t}}{-j2\pi f} \Big|_{-\tau/2}^{\tau/2} = A \frac{e^{-j\pi f \tau} - e^{j\pi f \tau}}{-j2\pi f} = A \frac{\sin(\pi f \tau)}{\pi f} \\ &= A\tau \cdot \text{sinc}(f\tau) \end{aligned}$$



Fourier Transform Properties



To help aid in finding Fourier transforms of various signals, there are properties similar to those for Fourier Series.

Property	Signal	Transform, Hz	Transform, rad/sec
	$x(t)$	$X(f)$	$X(\omega)$
	$y(t)$	$Y(f)$	$Y(\omega)$
Linearity	$ax(t) + by(t)$	$aX(f) + bY(f)$	$aX(\omega) + bY(\omega)$
Time Shift	$x(t - t_o)$	$X(f)e^{-j2\pi f t_o}$	$X(\omega)e^{-j\omega t_o}$
Freq. Shift	$x(t)e^{j2\pi f_o t}$	$X(f - f_o)$	$X(\omega - \omega_o)$
Modulation	$x(t)\cos(2\pi f_o t)$	$\frac{1}{2}X(f - f_o) + \frac{1}{2}X(f + f_o)$	$\frac{1}{2}X(\omega - \omega_o) + \frac{1}{2}X(\omega + \omega_o)$
Time Reversal	$x(-t)$	$X(-f)$	$X(-\omega)$
Scaling	$x(at)$	$\frac{1}{ a }X\left(\frac{f}{a}\right)$	$\frac{1}{ a }X\left(\frac{\omega}{a}\right)$
Conjugation	$x^*(t)$	$X^*(-f)$	$X^*(-\omega)$

Fourier Transform Properties



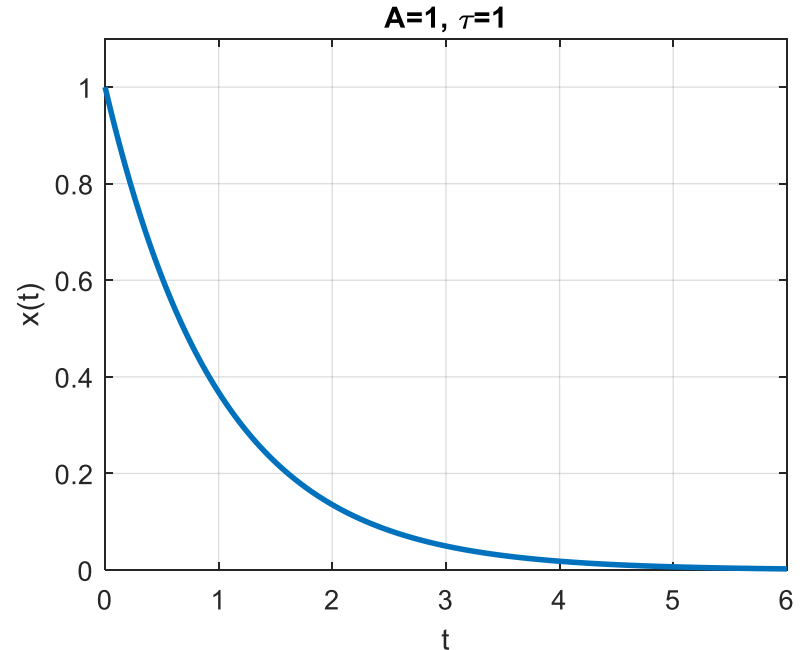
More properties

Property	Signal	Transform, Hz	Transform, rad/sec
	$x(t)$	$X(f)$	$X(\omega)$
	$y(t)$	$Y(f)$	$Y(\omega)$
Differentiation	$\frac{dx}{dt}$	$j2\pi f X(f)$	$j\omega X(\omega)$
Integration (assuming $X(0) = 0$)	$\int_{-\infty}^t x(u) du$	$\frac{X(f)}{j2\pi f}$	$\frac{X(\omega)}{j2\pi f}$
Multiplication	$x(t)y(t)$	$\int_{-\infty}^{\infty} X(u)Y(f-u) du$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} X(u)Y(\omega-u) du$
Convolution	$\int_{-\infty}^{\infty} x(\tau)y(t-\tau) d\tau$	$X(f) Y(f)$	$X(\omega) Y(\omega)$
Even Signals	$x(t)$ real and even	$X(f)$ real and even	$X(\omega)$ real and even
Odd Signals	$x(t)$ real and odd	$X(f)$ imaginary and odd	$X(\omega)$ imaginary and odd

Common Fourier Transforms



Example 11.1: Find the Fourier Transform for the one-sided exponential, $x(t) = Ae^{-t/\tau}u(t)$



Answer:

$$X(\omega) = \frac{A\tau}{1+j\omega\tau}$$

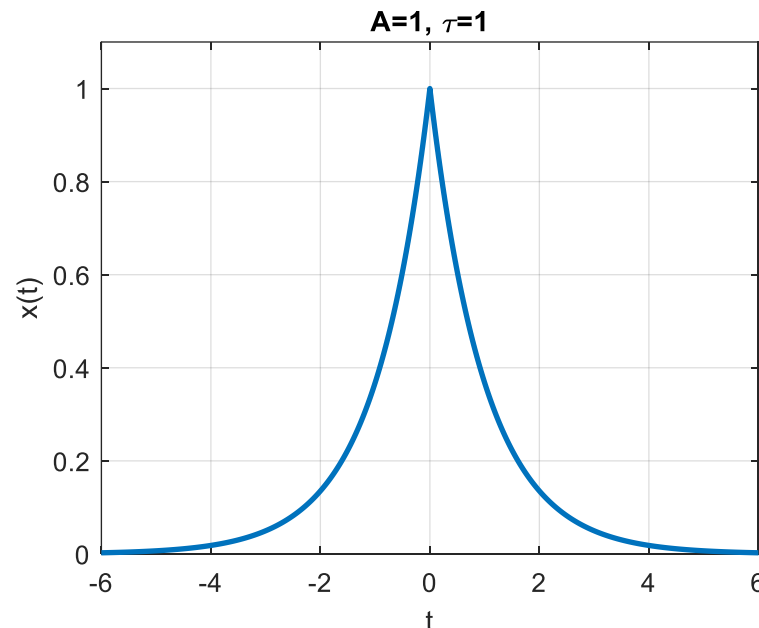
or

$$X(f) = \frac{A\tau}{1+j2\pi f\tau}$$

Common Fourier Transforms



Example 11.2: Find the Fourier Transform for the two-sided exponential, $y(t) = Ae^{-|t|/\tau}$



Answer:

$$Y(\omega) = \frac{2A\tau}{1+(\omega\tau)^2}$$

or

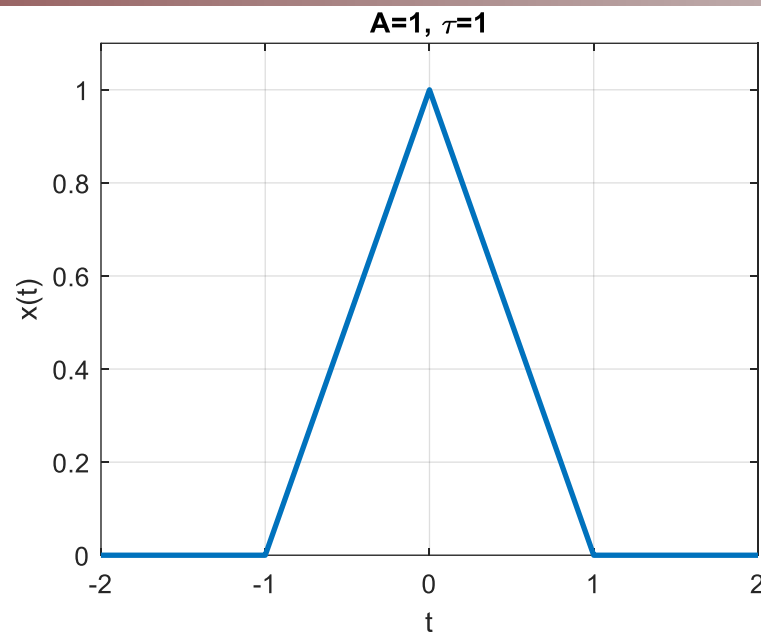
$$Y(f) = \frac{2A\tau}{1+(2\pi f\tau)^2}$$

Common Fourier Transforms



Example 11.3: Find the Fourier Transform for the triangle function,

$$w(t) = A \cdot \text{tri}\left(\frac{t}{\tau}\right) = \begin{cases} A \left(1 - \left|\frac{t}{\tau}\right|\right), & |t| < \tau, \\ 0, & |t| > \tau. \end{cases}$$



Answer:

$$W(\omega) = A\tau \text{sinc}^2\left(\frac{\omega\tau}{2\pi}\right)$$

or

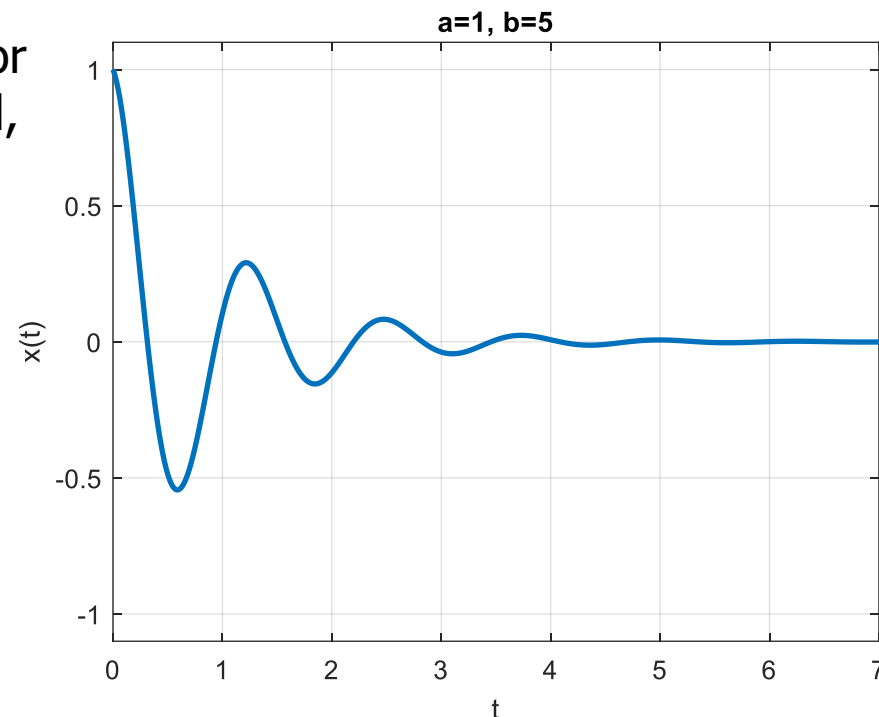
$$W(f) = A\tau \text{sinc}^2(f\tau)$$

Common Fourier Transforms



Example 11.4: Find the Fourier Transform for the one-sided exponentially decaying sinusoid,

$$z(t) = Ae^{-at}\cos(bt)u(t)$$



Answer:

$$Z(\omega) = \frac{A(a + j\omega)}{a^2 + b^2 - \omega^2 + 2ja\omega}$$



Fourier Transforms of One-Sided Signals



- For signals that are one-sided ($=0$ for $t < 0$), the Fourier transform is closely related to the Laplace transform.

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = \int_0^{\infty} x(t)e^{-j\omega t} dt = X(s)|_{s=j\omega}.$$

- We can use the table of Laplace transforms to find the Fourier transform of many one-sided signals.
- **Caution:** Only use entries from the table of Laplace transforms where the region of converge includes the imaginary axis ($s = j\omega$).

Example 11.5:

$$x(t) = Ae^{-at}\cos(bt)u(t)$$

$$X(s) = \frac{s+a}{(s+a)^2+b^2} \text{ for } \operatorname{Re}[s] > -a.$$

As long as $a > 0$, the ROC will include $s = j\omega$ (where $\operatorname{Re}[j\omega] = 0$).

$$\begin{aligned} X(\omega) &= \left. \frac{s+a}{(s+a)^2+b^2} \right|_{s=j\omega} \\ &= \frac{a+j\omega}{(a+j\omega)^2+b^2} \end{aligned}$$

Example 11.6:

$$x(t) = u(t)$$

$$X(s) = \frac{1}{s} \text{ for } \operatorname{Re}[s] > 0.$$

In this case, the ROC does not include $s = j\omega$ (where $\operatorname{Re}[j\omega] = 0$). Therefore

$$X(\omega) \neq \left. \frac{1}{s} \right|_{s=j\omega} = \frac{1}{j\omega}$$

Fourier Transforms of Common Signals



- Like we did with Laplace transforms, we can construct a table of Fourier transforms of some common signals.
- We'll add more to this table in the next lecture.

Signal	Transform, Hz, $X(f)$	Transform, rad/sec, $X(\omega)$
$\text{rect}(t)$	$\text{sinc}(f)$	$\text{sinc}\left(\frac{\omega}{2\pi}\right)$
$\text{sinc}(t)$	$\text{rect}(f)$	$\text{rect}\left(\frac{\omega}{2\pi}\right)$
$\text{tri}(t)$	$\text{sinc}^2(f)$	$\text{sinc}^2\left(\frac{\omega}{2\pi}\right)$
$\text{sinc}^2(t)$	$\text{tri}(f)$	$\text{tri}\left(\frac{\omega}{2\pi}\right)$
$e^{- t }$	$\frac{2}{1 + (2\pi f)^2}$	$\frac{2}{1 + \omega^2}$
$\frac{1}{1 + t^2}$	$\pi e^{- 2\pi f }$	$\pi e^{- \omega }$
$e^{-\pi t^2}$	$e^{-\pi f^2}$	$e^{-\omega^2/(4\pi)}$